

ASESII 24A02 Final Project - Part 2

2-way ANOVA method

Group 02

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1 Abstract

2 Dataset selection

2.1 Data Overview

```

data <- read.csv("data/penguins_size.csv", header = TRUE)

# Treat "." as a missing value
data[data == "."] <- NA

config <- list(
  file = "penguins_size.csv",
  response = "body_mass_g",
  factors = c("species", "sex")
)

data_raw <- data

analysis_data <- data_raw[
  ,
  c(config$response, config$factors),
  drop = FALSE
]

colSums(is.na(analysis_data))

```

```

## body_mass_g    species      sex
##           2           0      11

```

```

analysis_data <- na.omit(analysis_data)

if (anyNA(analysis_data)) {
  stop("Missing values remain in the analysis data.")
}

```

Table 1: Data Overview Before Missing-Value Removal

Variable	Role	Type	Distinct	Missing
species	Predictor	character	3	0
island	Predictor	character	3	0
culmen_length_mm	Predictor	numeric	164	2
culmen_depth_mm	Predictor	numeric	80	2
flipper_length_mm	Predictor	integer	55	2
body_mass_g	Response	integer	94	2
sex	Predictor	character	2	11

For 2-Way ANOVA, this method requires the dataset to have at least 2 categorical variables along with a continuous response variable for the analysis.

In general, from our chosen dataset, we decide to evaluate the factors that are associated with penguin body mass. This dataset contains information about 344 penguins, including species,

island, physical measurements such as culmen length, culmen depth, and flipper length, body mass, and sex. It is suitable for exploratory data analysis and 2-Way ANOVA analysis based on the biological and physical characteristics of penguins.

Specifically, the dataset contains 3 categorical variables: **species** (Adelie/Chinstrap/Gentoo), **island** (Torgersen/Biscoe/Dream), and **sex** (MALE/FEMALE). In this method, we select **species** and **sex** as the two categorical factors and **body_mass_g** as the continuous response variable. The analysis is used to test whether the two factors are associated with differences in mean body mass, as well as whether there is an interaction between the two factors.

2.2 Research Questions & Hypotheses

We investigate whether penguin species and sex are associated with differences in mean body mass, and whether there is an interaction between the two factors.

We follow the 2-way ANOVA model:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

with $\epsilon_{ijk} \sim N(0, \sigma^2)$.

- μ is the overall mean.
- α_i is the effect of the i -th level of Factor A, **species**.
- β_j is the effect of the j -th level of Factor B, **sex**.
- $(\alpha\beta)_{ij}$ is the interaction effect between Factor A and Factor B.
- ϵ_{ijk} is the random error term.

2.2.1 Factor A: Penguin Species

Research Question:

Is penguin species associated with differences in mean body mass?

$$\begin{cases} H_{0a} : \alpha_1 = \alpha_2 = \alpha_3 = 0 \\ H_{1a} : \text{Not all } \alpha_i \text{ are equal to 0.} \end{cases}$$

2.2.2 Factor B: Penguin Sex

Research Question:

Is the sex of penguin associated with differences in mean body mass?

$$\begin{cases} H_{0b} : \beta_1 = \beta_2 = 0 \\ H_{1b} : \text{Not all } \beta_j \text{ are equal to 0.} \end{cases}$$

2.2.3 Interaction Effect

Research Question:

Does the association between sex and penguin body mass depend on the penguin species?

$$\begin{cases} H_{0ab} : (\alpha\beta)_{ij} = 0 & \text{for all } i, j \\ H_{1ab} : \text{Not all } (\alpha\beta)_{ij} \text{ are equal to 0.} \end{cases}$$

2.3 Descriptive Statistics

2.3.1 Data Summary Statistics

Table 2: Overall Summary Statistics of Penguin Body Mass

N	Mean	Median	Variance	SD	Min	Max
333	4207.06	4050	648372.5	805.22	2700	6300

The dataset contains 333 observations of penguin body mass after removing observations with missing values. The mean body mass is 4207.06 g, with a median of 4050 g, indicating that the body masses are centered around approximately 4.0 to 4.2 kg. The standard deviation is 805.22 g, showing moderate variability in body mass, while the observed values range from 2700 g to 6300 g. The mean being slightly higher than the median suggests a mild right skew in the overall response distribution.

Table 3: Summary Statistics by Treatment Combination

Species	Sex	N	Mean	Median	Variance	SD	Min	Max
Adelie	FEMALE	73	3368.84	3400	72565.64	269.38	2850	3900
Adelie	MALE	73	4043.49	4000	120278.25	346.81	3325	4775
Chinstrap	FEMALE	34	3527.21	3550	81415.44	285.33	2700	4150
Chinstrap	MALE	34	3938.97	3950	131143.61	362.14	3250	4800
Gentoo	FEMALE	58	4679.74	4700	79286.34	281.58	3950	5200
Gentoo	MALE	61	5484.84	5500	98068.31	313.16	4750	6300

The dataset contains three levels of species and two levels of sex. Male penguins generally have higher mean body mass than female penguins across all three species.

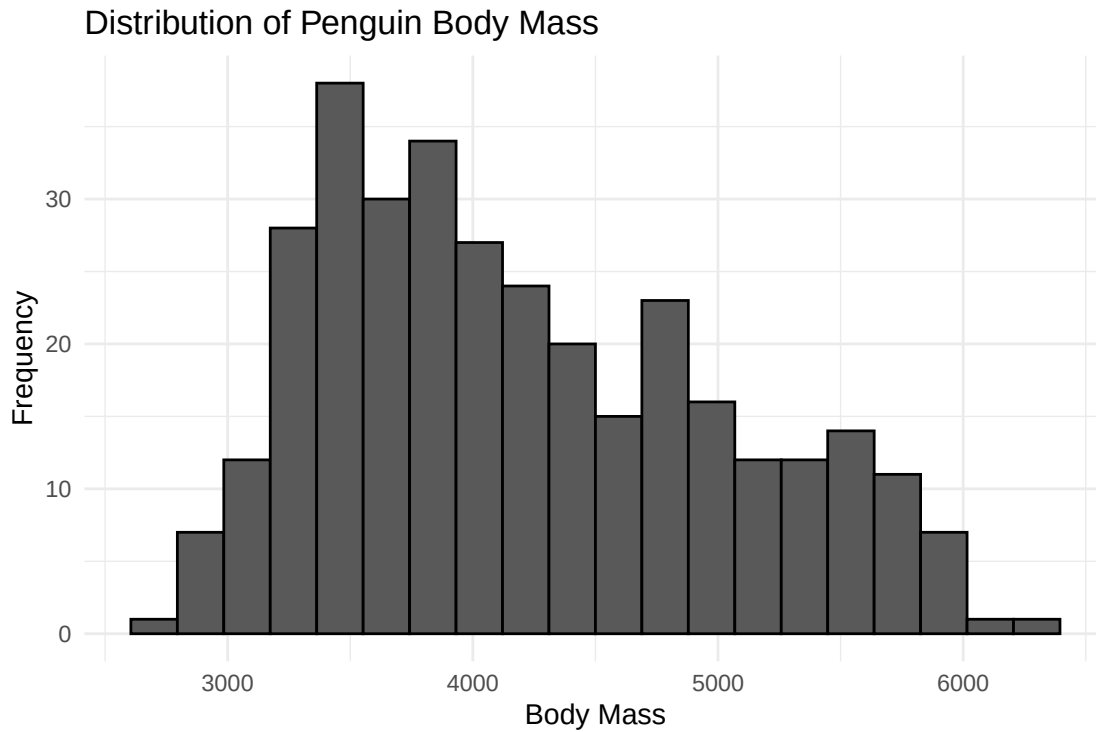
The difference in mean body mass between males and females varies across species. The difference is approximately 674.65 g for Adelie penguins, 411.76 g for Chinstrap penguins, and 805.10 g for Gentoo penguins. This variation provides descriptive evidence of a potential interaction between species and sex.

Among the six treatment combinations, the Gentoo male group has the highest mean body mass (5484.84 g), while the Adelie female group has the lowest mean body mass (3368.84 g). The group variances range from 72,565.64 to 120,278.25, with the Adelie male group having the largest

variance. Therefore, the homogeneity of variance assumption should be checked before conducting the two-way ANOVA.

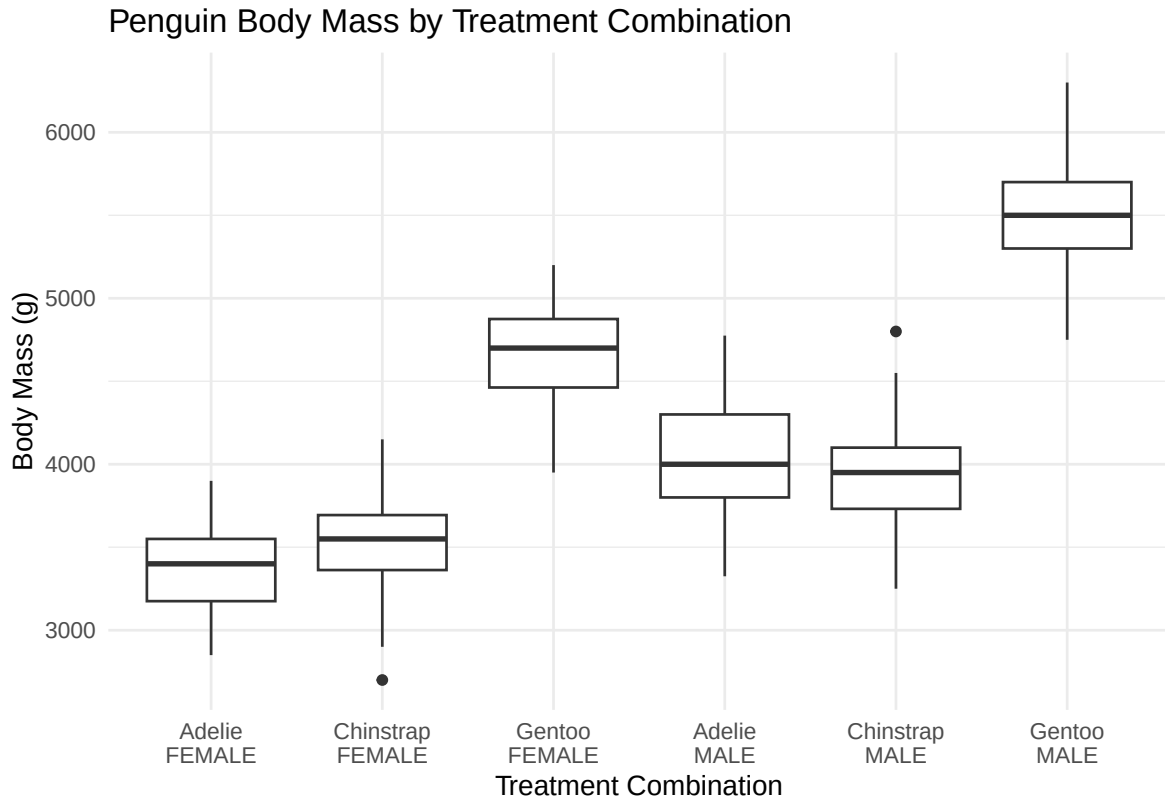
2.3.2 Visualization

2.3.2.1 Histogram of Penguin Body Mass



The histogram shows that penguin body masses range from approximately 2700 g to 6300 g, with most observations concentrated between 3000 g and 5000 g. The distribution shows a mild right skew, with a longer tail toward higher body masses. This is consistent with the mean body mass (4207.06 g) being higher than the median (4050 g). The distribution also shows relatively high frequencies around 3500 to 4000 g, while fewer observations are found at the upper end of the body mass range.

2.3.2.2 Box Plot

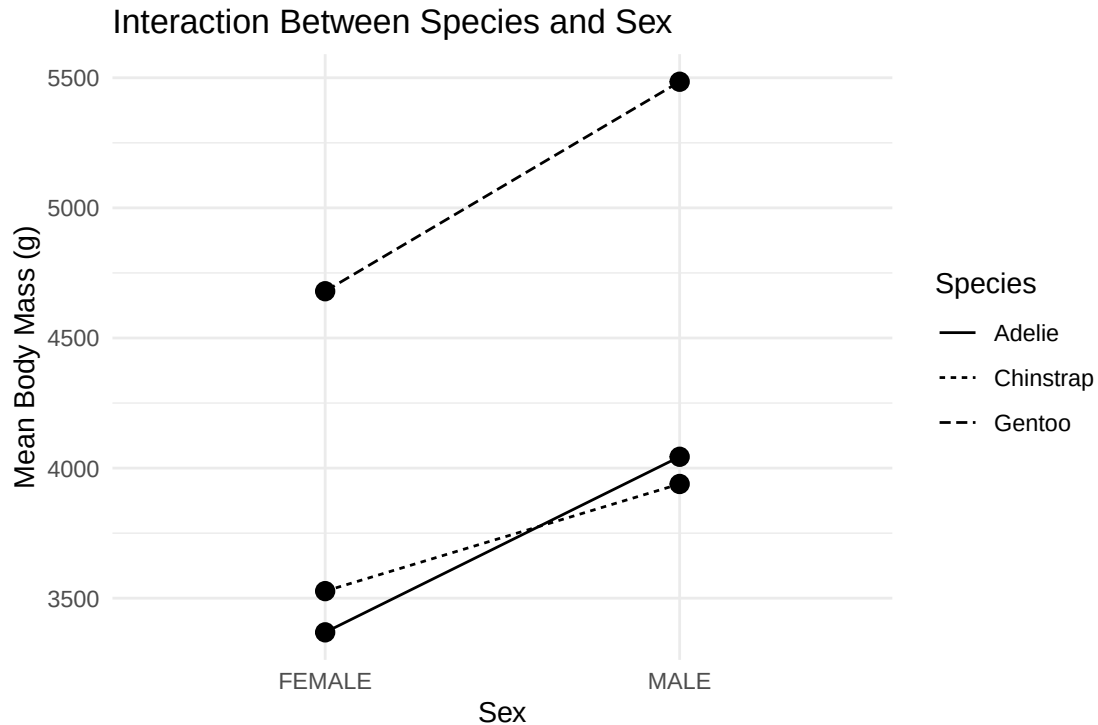


The box plot compares the distribution of penguin body mass across the six treatment combinations of species and sex. Male penguins generally have higher median and mean body mass than female penguins within each species.

The difference between male and female body mass appears across all three species. The largest difference is observed for Adeline penguins, with mean body mass increasing from 3368.84 g for females to 4043.49 g for males. For Chinstrap penguins, the corresponding means are 3527.21 g and 3938.97 g, while for Gentoo penguins they are 4679.74 g and 5484.84 g.

The spread of the groups is not completely uniform, with some groups showing wider variability than others. A few potential low outliers are also visible, particularly in the Chinstrap female group. Overall, the plot suggests that both species and sex may be associated with penguin body mass, while the differences in the male-female effect across species provide descriptive evidence of a possible interaction between the two factors.

2.3.2.3 Interaction Plot



The interaction plot shows the mean body mass for each combination of species and sex. Across all three species, male penguins have higher mean body mass than female penguins. The mean body mass increases by approximately 674.66 g for Adelie, 411.76 g for Chinstrap, and 805.10 g for Gentoo penguins.

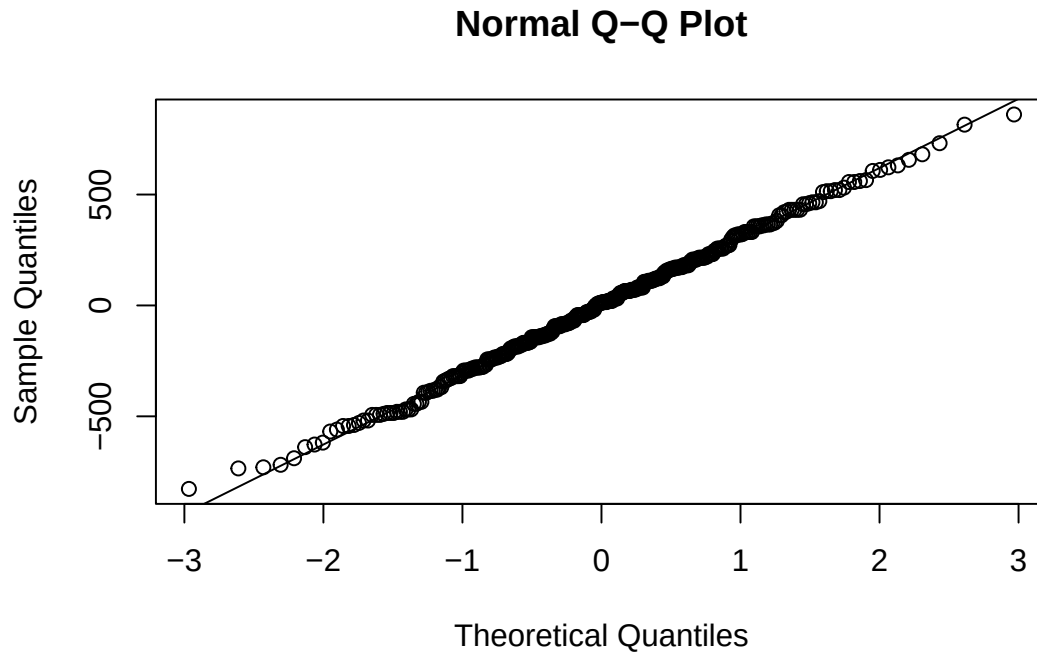
The lines are not perfectly parallel. In particular, the Adelie and Chinstrap lines cross between the female and male groups, while the Adelie and Gentoo lines show relatively similar slopes. This provides some descriptive evidence that the difference in body mass between males and females may vary across species.

However, the interaction pattern appears relatively modest overall, and the plot alone cannot determine whether the interaction is statistically significant. The significance of the species-by-sex interaction will be formally assessed using the two-way ANOVA.

2.3.3 Assumptions Check

```
qqnorm(residual)
qqline(residual)
```

2.3.3.1 Normality of Residuals



```
shapiro.test(residual)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residual
## W = 0.99776, p-value = 0.9367
```

The normality assumption was assessed using the normal Q-Q plot and the Shapiro-Wilk test. The hypotheses for the Shapiro-Wilk test are:

$$\begin{cases} H_0 : \text{The residuals follow a normal distribution} \\ H_1 : \text{The residuals do not follow a normal distribution} \end{cases}$$

The Shapiro-Wilk test resulted in

$$W = 0.99776, \quad p = 0.9367.$$

Since the p-value is greater than 0.05, we fail to reject H_0 . Therefore, there is no statistical evidence that the ANOVA residuals deviate from a normal distribution.

The Q-Q plot also shows that the residuals generally follow the reference line, with only minor deviations at the tails. The Shapiro-Wilk test gives $W = 0.99776$ with a p-value of 0.9367, providing further support for the normality assumption.

Overall, the residuals show no substantial departure from normality, and the normality assumption is considered satisfied for the subsequent two-way ANOVA.


```
leveneTest(
  body_mass_g ~
    species * sex,
  data = analysis_data
)
```

2.3.3.2 Homogeneity of Variances

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group      5  1.3908 0.2272
##           327
```

The homogeneity of variance assumption was assessed using the treatment group box plots and Levene's test. The six treatment combinations are formed by **species** and **sex**.

The hypotheses are:

$$\begin{cases} H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \dots = \sigma_6^2 \\ H_1 : \text{At least one treatment group has a different variance} \end{cases}$$

Levene's test produced

$$F(5, 327) = 1.3908, \quad p = 0.2272$$

Since the p-value is greater than 0.05, we fail to reject H_0 . Therefore, there is no statistical evidence that the variances differ across the six treatment combinations.

The group variances range from approximately 72,565.64 to 120,278.25, with the Adelie male group having the largest variance and the Adelie female group having the smallest variance. Although the group variances are not identical, Levene's test suggests that the differences are not statistically significant. Therefore, the homogeneity of variance assumption is considered satisfied for the subsequent two-way ANOVA.

```
sum(duplicated(analysis_data))
```

2.3.3.3 Independence of Errors

```
## [1] 169
```

The independence assumption requires that the error terms are independent:

$$\epsilon_{ijk} \stackrel{ind}{\sim} N(0, \sigma^2).$$

The independence assumption was assessed based on the structure of the dataset. Each observation represents an individual penguin, and the dataset does not contain repeated measurements or a time-dependent structure. Although some observations have identical values for the selected variables (`body_mass_g`, `species`, and `sex`), these do not necessarily represent duplicate penguins, since different penguins can naturally have the same body mass and characteristics.

Therefore, independence is considered reasonable based on the available dataset information. However, it cannot be formally verified because the dataset does not provide a unique identifier or detailed information about the sampling procedure.

2.3.4 Overall Assessment

The independence assumption is considered reasonable based on the structure of the dataset. Each observation represents an individual penguin, and there is no time-dependent or repeated-measurement structure identified in the analysis data. However, independence cannot be formally verified because the dataset does not provide a unique identifier or detailed information about the sampling procedure.

The normality assumption is considered satisfied based on both the Q-Q plot and the Shapiro-Wilk test ($p = 0.9367$), which provides no statistical evidence of a departure from normality. The homogeneity of variance assumption is also considered satisfied based on the non-significant Levene's test ($p = 0.2272$).

Overall, the assumptions are considered reasonably adequate for conducting the two-way ANOVA. No substantial concerns are observed regarding normality or homogeneity of variance, while the independence assumption is considered reasonable based on the available dataset information.

3 ANOVA / Factorial design analysis

```
model <- aov(body_mass_g ~ species * sex ,data=analysis_data)

anova_table <- as.data.frame(summary(model)[[1]])
p_value <- anova_table$`Pr(>F)`
f_value <- anova_table$`F value`

kable(anova_table, digits=3, caption = "ANOVA result")
```

Table 4: ANOVA result

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595109.56	758.358	0
sex	1	37090262	37090261.78	387.460	0
species:sex	2	1676557	838278.37	8.757	0
Residuals	327	31302628	95726.69	NA	NA

Using significant level α of 0.05, we observe that all these p-value are less than α .

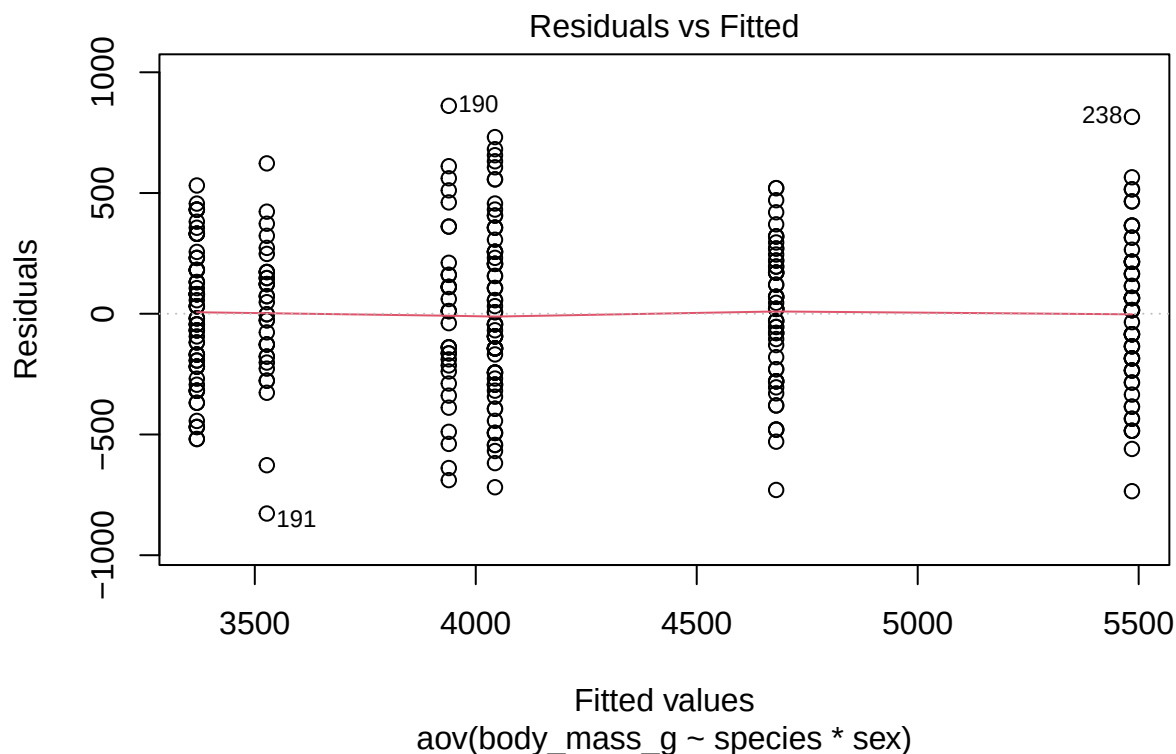
For species, the sum of squares is 145,190,219, which is the largest among all factors. Together with the very large F-value of 758.358 and $p < 0.001$, this indicates that species explains a very large amount of the variation in the response variable and has a highly significant effect.

For sex, the sum of squares is 37,090,262, which is also large compared with the residual sum of squares. The F-value of 387.460 and $p < 0.001$ show that sex has a highly significant effect on the response variable.

For the species - sex interaction, the degrees of freedom is 2, because the interaction involves the three species and two sex groups. Its sum of squares is 1,676,557, which is much smaller than the sums of squares for species and sex. However, the interaction is still statistically significant, with an F-value of 8.757 and $p < 0.001$. This indicates that although the interaction explains less variation, the effect of sex still differs significantly among species.

Overall, species contributes the largest amount of variation, as shown by its highest Sum Sq (145,190,219) and F-value (758.358). Sex also contributes a substantial amount of variation, with a Sum Sq of 37,090,262 and F-value of 387.460. The interaction has a much smaller Sum Sq (1,676,557) and F-value (8.757), but it remains significant. Therefore, all three effects—species, sex, and species x sex are statistically significant ($p < 0.001$), with species showing the strongest effect.

```
plot(model, which = 1)
```

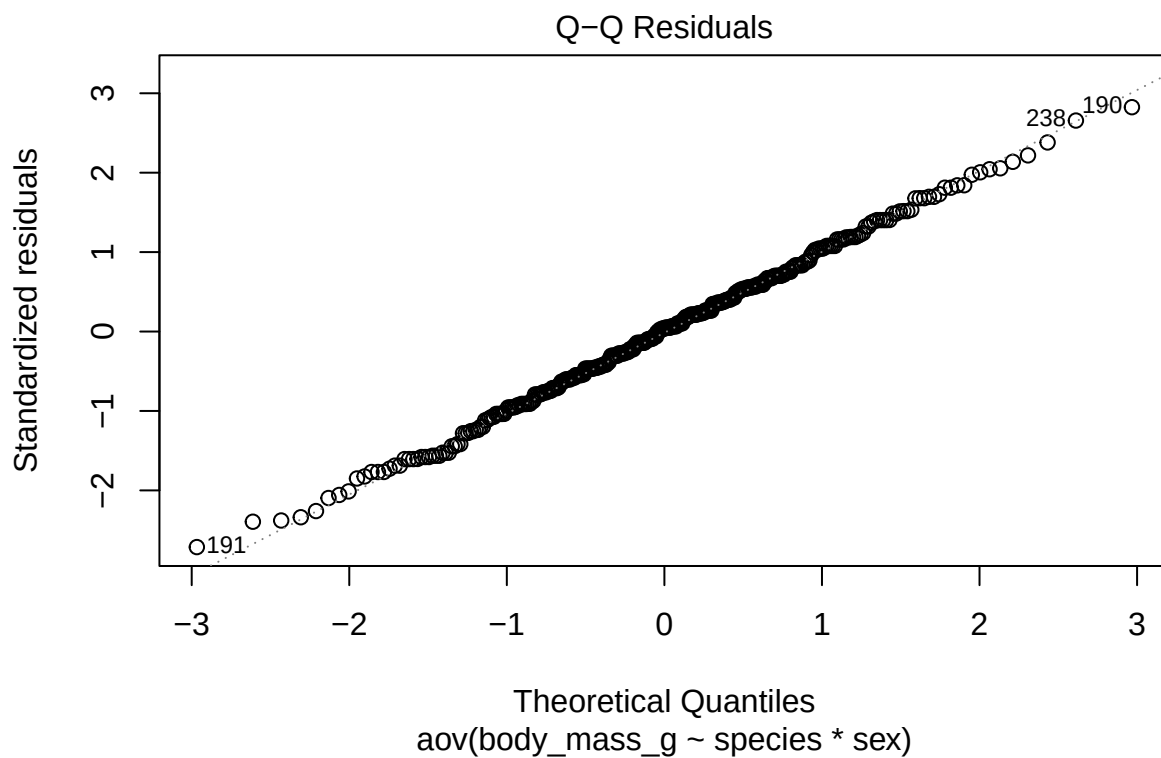


This residuals are generally scattered around the horizontal zero line across the different fitted

values, and the red trend line remains close to zero. This indicates that there is no strong systematic pattern or curvature in the residuals, suggesting that the ANOVA model captures the main structure of the data reasonably well.

The spread of the residuals is somewhat different between the fitted-value groups, but there is no clear funnel-shaped pattern where the spread consistently increases or decreases as the fitted values increase. Therefore, there is no strong visual evidence of severe heteroscedasticity from this plot.

```
plot(model, which = 2)
```



This Q-Q plot shows most of the points closely follow the diagonal reference line, particularly in the middle section of the plot, indicating that the residuals are approximately normally distributed. This suggests that the assumption of normality required for ANOVA is reasonably satisfied.

4 Posthoc analysis:

```
tukey <- TukeyHSD(model, which = "species")  
kable(tukey$species, digits = 3, caption = "Tukey HSD Results")
```

Table 5: Tukey HSD Results

	diff	lwr	upr	p adj
Chinstrap-Adelie	26.924	-80.026	133.874	0.824
Gentoo-Adelie	1386.273	1296.307	1476.238	0.000
Gentoo-Chinstrap	1359.349	1248.611	1470.087	0.000

The comparison between Chinstrap and Adelie is not statistically significant, with an adjusted p-value of 0.824, which is greater than 0.05. The confidence interval also includes zero (-80.026 to 133.874). Therefore, there is insufficient evidence to conclude that the mean response differs between Chinstrap and Adelie.

In contrast, Gentoo differs significantly from Adelie, with a mean difference of 1386.273 and an adjusted p-value less than 0.001. The 95% confidence interval (1296.307 to 1476.238) does not contain zero, providing strong evidence that the two species have different mean responses. Since the mean difference is positive, the mean response for Gentoo is higher than Adelie by approximately 1386.273 units.

Similarly, Gentoo differs significantly from Chinstrap, with a mean difference of 1359.349 and an adjusted p-value less than 0.001. The 95% confidence interval (1248.611 to 1470.087) is entirely positive, indicating that the mean response for Gentoo is higher than Chinstrap by approximately 1359.349 units.

Overall, the Tukey HSD results show that Gentoo is significantly different from both Adelie and Chinstrap, whereas Adelie and Chinstrap do not differ significantly from each other. This provides more specific information about the significant species effect identified by the ANOVA.

```
sex_analysis <- t.test(body_mass_g ~ sex, data=analysis_data)

tidy_t_test <- tidy(sex_analysis)

kable(
  tidy_t_test[, c("estimate", "statistic", "p.value", "conf.low", "conf.high")],
  digits = 3,
  col.names = c("Mean Difference", "t-Statistic", "p-value", "Lower CI", "Upper
    ↪ CI"),
  caption = "Independent t-test Comparing Penguin Body Mass by Sex"
)
```

Table 6: Independent t-test Comparing Penguin Body Mass by Sex

Mean Difference	t-Statistic	p-value	Lower CI	Upper CI
-683.412	-8.555	0	-840.578	-526.245

An independent-samples t-test was conducted to compare penguin body mass between male and female penguins. The results showed a mean difference of -683.412 grams, with a t-statistic of -8.555 and a p-value of < 0.001 .

The very small p-value indicates a statistically significant difference in body mass between the two sexes. The mean difference of -683.412 grams indicates that the first sex group in the comparison has an average body mass approximately 683.4 grams lower than the second sex group.

The 95% confidence interval for the mean difference ranges from -840.578 to -526.245 grams. Since the entire interval is below zero, the estimated difference is consistently in the same direction. Therefore, the results indicate that one sex has substantially greater average body mass than the other.

Overall, sex is associated with a significant difference in penguin body mass, with an average difference of approximately 683 grams between the two sexes.

```
emm_result <- emmeans(model, pairwise ~ sex | species, adjust = "tukey")

kable(
  as.data.frame(emm_result$contrasts),
  digits = 3,
  col.names = c("Contrast", "Species", "Difference (g)", "Std. Error", "df",
    ↪ "t-ratio", "p.value"),
  align = "llrrrrr",
  caption = "Pairwise Comparisons of Body Mass Between Sexes Stratified by
    ↪ Penguin Species"
)
```

Table 7: Pairwise Comparisons of Body Mass Between Sexes
Stratified by Penguin Species

Contrast	Species	Difference (g)	Std. Error	df	t-ratio	p.value
FEMALE - MALE	Adelie	-674.658	51.212	327	-13.174	0
FEMALE - MALE	Chinstrap	-411.765	75.040	327	-5.487	0
FEMALE - MALE	Gentoo	-805.095	56.743	327	-14.188	0

Pairwise comparisons showed that male penguins had significantly higher body mass than females in all three species ($p < 0.001$). The differences were approximately 674.7 g for Adelie, 411.8 g for Chinstrap, and 805.1 g for Gentoo. The largest sex difference occurred in Gentoo, while the smallest occurred in Chinstrap.

4.1 Practical significance

The results are not only statistically significant but also practically meaningful. The differences in body mass between species are large, particularly between Gentoo and the other two species, with differences of more than 1,300 g. The sex differences are also substantial, ranging from approximately 412 g to 805 g, with males consistently heavier than females.

Therefore, species and sex should be considered when assessing or comparing penguin body mass. Ignoring these factors could lead to misleading comparisons, especially when comparing groups with different species or sex compositions.

5 Conclusion and evaluation

The two-way ANOVA showed that species and sex both have significant effects on penguin body mass ($p < 0.001$). The Species x Sex interaction was also significant ($p < 0.001$), indicating that the effect of sex on body mass varies among species. Tukey's HSD showed that Gentoo differs significantly from both Adelie and Chinstrap, while Adelie and Chinstrap do not differ significantly. Pairwise comparisons further showed that males have significantly higher body mass than females in all three species, with the largest difference observed in Gentoo.

The analysis provides strong evidence of differences in body mass by species and sex. However, the ANOVA assumptions should be considered, particularly normality and homogeneity of variance. The data may also be unbalanced across groups, which can affect the analysis. Future analysis could use robust methods or transformations if assumption violations are substantial.

Based on these results, species and sex should both be considered when analyzing or predicting penguin body mass. In particular, the significant interaction suggests that the effect of sex should not be interpreted independently of species. Further studies could collect more balanced samples across species and sexes to improve the reliability and generalizability of the findings.